

# Adviser Review Memo

Semiparametric Estimation of Myerson Reserve Prices with WGAN-GP and DML

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## Decision requested

I would like guidance on whether the current methodological and Monte Carlo paper is ready for a computational-econometrics submission, or whether an empirical auction application should be treated as a prerequisite for an upward journal target. I am not seeking another round of estimator tuning before this decision.

## Paper in one sentence

When only auction maxima are observed, a structural WGAN can learn a coherent global latent value distribution, but valid reserve-price inference requires separate control of local CDF and density moments; cross-fitting and orthogonality do not compensate for missing local identifying information.

## What the paper contributes

- It embeds the auction order-statistic observation rule inside a bidder-level WGAN-GP, preserving symmetric independent private values before taking the observed maximum.
- It derives a cross-fitted Neyman-orthogonal score and an asymptotic linearity result for a fixed-bandwidth regularized reserve.
- It proves the required local nuisance rate for a training-fold empirical kernel learner under primitive fixed-bandwidth and local-overlap conditions.
- Nested Monte Carlo ablations isolate the failure of WGAN-based inference in local tail moments and motivate a hybrid global-WGAN/local-score estimator.
- External-validity designs identify a local-information boundary: the formal procedure is stable with three or five bidders but deteriorates with ten bidders when maxima rarely occur near the reserve.

## Claim boundary

**Formal claim:** valid inference for the fixed-bandwidth regularized reserve when the stated nuisance-rate, overlap, and root conditions hold; these conditions are verified for the empirical-local learner.

**Diagnostic evidence:** WGAN-GP global fit does not imply the local rate; Richardson correction, residual-bias envelopes, and root-union intervals for the exact Myerson reserve are finite-sample prototypes, not a uniform coverage theorem.

## Key evidence

- Replacing WGAN score nuisances with training-fold empirical-local moments restores near-oracle RMSE and coverage in the baseline design.
- Fixed-bandwidth target coverage is 92.0–96.5 percent across baseline, heavy-tail, and mixture distributions with three or five bidders.
- With ten bidders and weak local support, coverage falls to roughly 78 percent and multiple roots emerge.
- The stable  $n^{-0.2}$  exact-target prototype reaches 96.5 percent coverage after a residual-bias envelope, but remains explicitly diagnostic.

## Assumptions that should remain visible

Symmetric independent private values; observed auction maxima; known bidder count in the core design; local overlap around the reserve; and regular score root/Jacobian behavior. The paper should not imply that orthogonality repairs violations of these identifying conditions.

## Three questions for the review

1. Is the local-information boundary plus the hybrid estimator a clear enough econometric contribution, or does the paper still read mainly as a WGAN application?
2. Is IJCEE a sensible current-form target, or is the expected return high enough to first add an empirical bridge and target *The Econometrics Journal* or *Journal of Econometric Methods*?
3. Which maintained auction assumption is most likely to concern a referee: symmetry/IPV, known bidder count, maxima-only observation, or local overlap?

## Suggested reading order

Abstract and Introduction; Estimation and Inference; the local-nuisance and external-validity simulation results; Conclusion; then the proof appendix as needed. The identified and anonymous manuscripts, replication guide, and archived results are synchronized in the project repository.